STACKS, SHEAVES ON ALGEBRAIC STACKS: LEMMA-LOCALLY-QUASI-COHERENT-ALTERNATIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 4798-4816 of sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex; proof span is lines 4818-4842, the complete author proof (23 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 19 of queue_algebraic-stacks-sheaves.json (pb=1296, stmt 4798-4816, proof 4818-4842).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0GZX: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4842–4842. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma equation-alternative-ringed: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4727–4731. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma section-restriction: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 1185–1200. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 021B: sources/stacks-algebraic-stacks-sheaves/pinned-original-source.tex, source lines 4842–4842. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $p : \mathcal{X} \to (Sch/S)_{fppf}$ be a category fibred in groupoids. Let $\mathcal{F}$ be an $\mathcal{O}$-module on $\mathcal{X}_{affine}$. The following are equivalent*

(1) *for every morphism $x \to x'$ of $\mathcal{X}_{affine}$ such that $p(x) \to p(x')$ is an étale morphism (of affine schemes), the map $\mathcal{F}(x') \otimes_{\mathcal{O}(x')} \mathcal{O}(x) \to \mathcal{F}(x)$ is an isomorphism,*

Date: stacks-project:stacks-sheaves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) *$\mathcal{F}$ is a sheaf for the étale topology on $\mathcal{X}_{affine}$ and for every object x of $\mathcal{X}_{affine}$ the restriction $x^*\mathcal{F}|_{U_{affine,étale}}$ is quasi-coherent where $U = p(x)$,*
(3) *$\mathcal{F}$ corresponds to a locally quasi-coherent module on $\mathcal{X}$ via the equivalence (equation-alternative-ringed) for the étale topology.*

Proof. To make sense out of condition (2), recall that $U_{affine,étale}$ is the full subcategory of $U_{étale}$ consisting of affine objects, see Topologies, Definition 021B. Similarly to the discussion in Section section-restriction the object x of $\mathcal{X}_{affine}$ induces an equivalence $\mathcal{X}_{affine,étale}/x \to (Aff/U)_{étale}$ of sites. Then $x^*\mathcal{F}$ is the sheaf of modules on $(Aff/U)_{étale}$ corresponding to the restriction $\mathcal{F}|_{\mathcal{X}_{affine,étale}/x}$. Finally, using the continuous and cocontinuous inclusion functor $U_{affine,étale} \to (Aff/U)_{étale}$ we can further restrict and obtain $x^*\mathcal{F}|_{U_{affine,étale}}$.

The equivalence of (1) and (2) follows from the remarks above and Descent, Lemma 0GZX applied to the restriction of $\mathcal{F}$ to $U_{affine,étale}$ for every object x of $\mathcal{X}$ lying over an affine scheme U. The equivalence of (2) and (3) is immediate from the definitions and the fact that quasi-coherent modules on $U_{affine,étale}$ and $U_{étale}$ correspond (again by Descent, Lemma 0GZX for example). □